

Portfolio optimization with transaction costs: a two-period mean-variance model

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Abstract In this paper, we study a multiperiod mean-variance portfolio optimization problem in the presence of proportional transaction costs. Many existing studies have shown that transaction costs can significantly affect investors' behavior. However, even under simple assumptions, closed-form solutions are not easy to obtain when transaction costs are considered. As a result, they are often ignored in multiperiod portfolio analysis, which leads to suboptimal solutions. To provide better insight for this complex problem, this paper studies a two-period problem that considers one risk-free and one risky asset. Whenever there is a trade after the initial asset allocation, the investor incurs a linear transaction cost. Through a mean-variance model, we derive the closed-form expressions of the optimal thresholds for investors to re-allocate their resources. These thresholds divide the action space into three regions. Some important properties of the analytical solution are identified, which shed light on solving multiperiod problems.

Keywords Investment analysis · Multiperiod portfolio optimization · Mean-variance analysis · Transaction costs

1 Introduction

Portfolio selection is to optimize an asset combination according to the investor's preference on reward and risk. Transaction costs are fees relating to investment activities including commission, bid-ask spreads and so on. Such costs are not uncommon in practice but often ignored in research. In this section, we review portfolio optimization studies, focusing on the issue of transaction costs.

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1.1 Portfolio optimization without transaction costs

The way for modern portfolio selection theory has been paved by Markowitz (1952) using mean and variance as the measures of reward and risk of the portfolio respectively. Such a mean-variance analysis is the first effective approach to treat the trade-off between reward and risk quantitatively. For decades, the mean-variance approach has received great attention, and intensive research has been done in this area. Steinbach (2001) gives an extensive literature review of this approach.

Portfolio optimization can be studied by single-period models and long-term models. To distinguish from single-period models, long-term portfolio optimization models defined in this paper allow for interim rebalancing of the portfolio.

1.1.1 Single-period mean-variance portfolio selection without transaction costs

In a single-period problem, an investor constructs a portfolio by selecting a number of assets and allocating a portion of her wealth to each asset selected at the beginning of the period, based on observation of the market and anticipation of the performance of financial assets. Suppose there are N assets in the market, and the investor with an initial wealth of w_0 wants to allocate her wealth to the N assets available. Let $\mathbf{x}$ be the portfolio vector which indicates the amount of wealth allocated to each asset, $\mathbf{R} = (R_1, R_2, \dots, R_N)'$ be the random return rate vector with expectation $\bar{\mathbf{R}} = (E(R_1), E(R_2), \dots, E(R_N))'$, and $\mathbf{V}$ be the covariance matrix of $\mathbf{R}$.

The investor's expected wealth at the end of the period is $\bar{\mathbf{R}}\mathbf{x}$, and the variance is $\mathbf{x}'\mathbf{V}\mathbf{x}$. The portfolio is evaluated by its reward and risk, represented by mean and variance respectively. The investor can choose either to maximize her reward within a certain risk level σ , or to minimize the risk of the portfolio given a desired reward μ .

There exist various mean-variance formulations. A commonly used formulation maximizes the expected value of a concave quadratic utility. It takes the following form:

$$\begin{aligned} \max \quad & \bar{\mathbf{R}}\mathbf{x} - \lambda \mathbf{x}'\mathbf{V}\mathbf{x} \\ \text{s.t.} \quad & \mathbf{1}'\mathbf{x} = w_0 \\ & \mathbf{x} \geq \mathbf{0} \end{aligned} \quad (1)$$

where $\mathbf{1}$ is a vector of all ones, $\mathbf{0}$ is a vector of all zeros, and $\lambda > 0$ is a parameter reflecting the investor's risk tolerance. A higher value of λ indicates a stronger aversion towards investment risk. The Markowitz efficient frontier can be generated by solving (1) in terms of λ . Throughout this paper, our analysis is based on a similar mean-variance framework but with transaction costs.

1.1.2 Long-term portfolio optimization without transaction costs

The mean-variance framework has been widely used to solve single-period problems. However, this method has gained relatively less success when applied to long-term investment analysis. This is partly due to the aggregated uncertainty when long-term investment is considered (Li and Ng 2000). Difficulties of extending the single-period mean-variance model to a multiperiod scenario have been reported by Chen et al. (1971), Li and Ng (2000) and Steinbach (2001).

Long-term problems can be classified into two categories, namely discrete-time models and continuous-time models. In discrete-time multiperiod models, portfolio rebalancing

can be conducted at the beginning of each period. [Hakansson \(1971\)](#) models the discrete-time multiperiod investment problem with the objective of maximizing the average return rather than optimizing the tradeoff between mean and variance of the portfolio. [Gülpınar and Rustem \(2007\)](#) measure risk by the downside volatility instead of variance. [Osorio et al. \(2008\)](#) develop a framework for testing solutions to the multiperiod mean-variance investment problems considering tax. [Marzban et al. \(2013\)](#) apply robust analysis to a multiperiod problem considering stocks and options. [Li et al. \(1998\)](#) and [Li and Ng \(2000\)](#) embed the multiperiod mean-variance model into an auxiliary quadratic model. The solution to the auxiliary model was then used to derive the solution to the original mean-variance model. [Çelikyurt and Özekici \(2007\)](#) adopt the same method to an investment problem with the assumption that the market moves according to a Markov chain. However, in all these works, no transaction costs are considered in the formulation. This will be addressed at the beginning of the next section.

The continuous-time portfolio optimization problems assume that the portfolio can be rebalanced at any time. Instead of using mean-variance objectives, most continuous-time models maximize the expected discounted utility of consumption over a time horizon or the investor's wealth at the end of the investment horizon. [Samuelson \(1969\)](#) discusses a lifetime portfolio optimization under deterministic return rates of assets. [Merton \(1971\)](#), known as the pioneering work in continuous-time investment planning, develops the optimal investment policies under the assumptions that the utility functions belong to HARA (hyperbolic absolute risk-aversion) family and the asset price processes follow a geometric Brownian motion. [Cox and Huang \(1989\)](#) later improve the model by imposing nonnegativity constraints on the investor's consumption amount and final wealth. [Bi et al. \(2014\)](#) investigate the optimal strategy for the continuous-time investment problem under the no-bankruptcy constraint. The optimal consumption and investment strategies are derived by a technique using martingales. These works adopt an objective function maximizing the cumulative discounted utility of consumption or final wealth. There have been enormous amounts of literature studying continuous-time approaches for portfolio selection problems since then. As discussed by [Merton \(1971\)](#), the main advantage of the continuous-time approach is that it makes use of the established research in stochastic processes and reduces the number of parameters in the model. Nevertheless, discrete-time models are still important in practice. They are easier to understand and implement. Besides, it may not be optimal to have infinitesimal trading and we can therefore shorten the period of discrete-time models to approximate the performance of the continuous-time models.

1.2 Portfolio selection with transaction costs

It is reported that transaction costs can significantly affect investors' behaviors. [Constantinides \(1979\)](#) shows that transaction costs lead to less frequent trading. [Pelsser and Vorst \(1996\)](#) prove that transaction fees as modest as 0.5 % of the trading amount can make the optimal strategy of considering no transaction costs inferior to simple stop-loss and lock-in strategies. However, incorporating such transaction costs brings challenges in obtaining closed-form solutions. In this section, we review portfolio selection models with transaction costs for both single-period and multiperiod settings.

1.2.1 Single-period portfolio optimization with transaction costs

Single-period portfolio optimization with transaction costs is first studied by [Constantinides \(1979\)](#). [Perold \(1984\)](#) later incorporates concave piecewise linear transaction costs into a one-

period mean-variance portfolio revision model. This methodology is later adopted by [Konno and Wijayanayake \(2001\)](#) to analyze a mean-absolute-deviation model. [Best and Hlouskova \(2005\)](#) develop an efficient algorithm that solves a large-scale single-period investment problem with proportional transaction costs. [Lobo et al. \(2007\)](#) obtain approximate results for a one-period rebalancing problem with concave transaction costs. [Kozhan and Schmid \(2009\)](#) study an investment problem with transaction costs where the investment risk is determined based on subjective beliefs instead of using any probability distribution. [Jana et al. \(2009\)](#) incorporate proportional transaction costs into a mean-variance model with an additional diversification criterion. [Zhang et al. \(2010\)](#) consider transaction costs in a one-period rebalancing model treating the random return rates as fuzzy numbers. Although single-period models involving multiple assets or various transaction costs have been developed, their results cannot easily apply to multiperiod scenarios as the complexity of the problem grows rapidly with increasing number of investment periods.

1.2.2 Long-term portfolio optimization with transaction costs

Recently, more attention has been placed on long-term portfolio optimization in the presence of transaction costs. In terms of continuous-time problems, most existing models focus on maximizing the discounted utility of consumption or bequest instead of studying the trade-off between reward and risk as in the mean-variance models. [Davis and Norman \(1990\)](#) obtain closed-form solutions for a problem that maximizes the cumulative utility of consumption on an infinite horizon under proportional transaction costs. [Dumas and Luciano \(1991\)](#) solve a similar model with an objective to increase the final utility. [Liu \(2004\)](#) further incorporates fixed transaction costs into the continuous-time model, but no closed-form transaction thresholds are identified.

As for discrete-time models, [Bertsimas and Pachamanova \(2008\)](#) solve the corresponding problem by maximizing the worst-case return to get a robust solution instead of providing optimal solutions according to the investor's risk tolerance. This robust design offers a safe investment option but it is conservative and restricts the risk-taking investors from seeking higher returns. [Gennotte and Jung \(1994\)](#) adapt the model of [Dumas and Luciano \(1991\)](#) for continuous-time analysis to solve a discrete-time multiperiod problem. It maximizes a power utility function but no closed-form solution is obtained. [Chryssikou \(1998\)](#) models the multiperiod portfolio optimization problems with transaction costs under the mean-variance framework and obtains near-optimal solutions. However, he assumes that the transaction costs are quadratic functions of the trading amounts, which means that the unit transaction costs increase as the trading amounts go up. Such cost schemes are rarely seen in practice. To our best knowledge, no closed-form solution for the multiperiod mean-variance portfolio optimization problem with linear transaction costs has been obtained.

1.3 Overview

This paper aims to study the discrete-time multiperiod mean-variance portfolio optimization problem with transaction costs. The rest of the paper is organized as follows. In Sect. 2, we describe the mean-variance model of portfolio optimization in the presence of transactions costs and consider the problem involving only two assets, one risky and one risk-free. We also show that the mean-variance model of portfolio selection with transaction costs can be embedded in a quadratic model. In Sect. 3, we consider a two-period investment problem and formulate it as a dynamic program. The analytical solution for the two-period problem is also obtained. Section 4 gives a numerical example to illustrate the methodology, and

the mean-variance efficient frontier has been plotted for different transaction fee schedules. Lastly, some conclusions are drawn in Sect. 5.

2 Multiperiod mean-variance formulation with transaction costs

Suppose there are T periods and $N + 1$ assets. Here, asset 0 is a risk-free asset with a fixed return rate r^0 such as a cash account or a bond, while asset i is a risky asset with a random return rate $\hat{R}_t^i$ for time period t , where $i = 1, 2, \dots, N$ and $t = 0, 1, 2, \dots, T - 1$. Depending on the holding of asset i ($i = 0, 1, 2, \dots, N$) at the end of period t , which is denoted by h_t^i , the investor chooses the transaction amount of asset i to rebalance the asset holdings. The amount of risky asset i bought and sold at the end of time t are denoted by nonnegative variables x_t^{i+} and x_t^{i-} respectively. All the money that is not invested in the risky asset goes to the risk-free account. Once a transaction occurs on asset i , the investor has to pay a proportional transaction cost from the risk-free account at a rate of α^i for buying and β^i for selling. Thus the buying and selling costs are $\alpha^i x_t^{i+}$ and $\beta^i x_t^{i-}$ respectively. We first derive the recursive rebalancing equations.

2.1 Recursive rebalancing equations

At the beginning of each period, the investor can change her portfolio by trading the risky asset. The incurred transaction costs are deducted from the risk-free asset. The recursive rebalancing equations are thus

$$h_{t+1}^0 = r^0 \left(h_t^0 - \sum_{i=1}^N (1 + \alpha^i) x_t^{i+} + \sum_{i=1}^N (1 - \beta^i) x_t^{i-} \right) \quad (2)$$

$$h_{t+1}^i = \hat{R}_t^i \left(h_t^i + x_t^{i+} - x_t^{i-} \right) \quad (3)$$

for the risk-free asset and risky assets respectively.

These recursive rebalancing equations reflect how the wealth is accumulated through each time period. The multiperiod mean-variance portfolio optimization problem can then be formulated as

$$\begin{aligned} \text{PB}[\hat{\lambda}] : \quad & \max E(w_T) - \lambda \text{Var}(w_T) \\ & s.t. \quad w_T = h_T^0 + \sum_{i=1}^N (1 - \beta^i) h_T^i \\ & \quad h_{t+1}^0 = r^0 \left(h_t^0 - \sum_{i=1}^N (1 + \alpha^i) x_t^{i+} + \sum_{i=1}^N (1 - \beta^i) x_t^{i-} \right) \\ & \quad h_{t+1}^i = \hat{R}_t^i \left(h_t^i + x_t^{i+} - x_t^{i-} \right) \\ & \quad x_t^{i+}, x_t^{i-} \geq 0. \end{aligned} \quad (4)$$

for $i = 1, 2, \dots, N$ and $t = 0, 1, \dots, T - 1$. The final wealth w_T is the sum of the risk-free asset and the risky asset after liquidation. For notational simplicity, denote

$$\begin{aligned} R_t^i &= \hat{R}_t^i \quad \text{for } t = 0, 1, \dots, T - 2 \\ R_{T-1}^i &= \hat{R}_{T-1}^i (1 - \beta^i). \end{aligned} \quad (5)$$

Note that if liquidation is not required at the end of the investment horizon, then $R_{T-1}^i = \dot{R}_{T-1}^i$. Now $PB[\lambda]$ can be rewritten as

$$\begin{aligned} PB[\lambda] : \quad & \max E(w_T) - \lambda \text{Var}(w_T) \\ \text{s.t. } & w_T = h_T^0 + \sum_{i=1}^N h_T^i \\ & h_{t+1}^0 = r^0 \left(h_t^0 - \sum_{i=1}^N (1 + \alpha^i) x_t^{i+} + \sum_{i=1}^N (1 - \beta^i) x_t^{i-} \right) \\ & h_{t+1}^i = R_t^i \left(h_t^i + x_t^{i+} - x_t^{i-} \right) \\ & x_t^{i+}, x_t^{i-} \geq 0. \end{aligned} \quad (6)$$

Since the variance term cannot be decomposed with respect to time stages, $PB[\lambda]$ cannot be solved directly using dynamic programming. In the following sections, we obtain the optimal solutions by solving an auxiliary model.

2.2 Auxiliary model

It has been shown in Li and Ng (2000) that maximizing the mean-variance objective function of $PB[\lambda]$ is equivalent to maximizing the following quadratic function:

$$\delta E(w_T) - \lambda E(w_T)^2 \quad (7)$$

when δ takes the value of $1 + 2\lambda E(w_T)|_{\pi^*}$, where π^* is the optimal investment policy.

Therefore, the solutions to the mean-variance problem $PB[\lambda]$ can be generated by solving the following auxiliary problem $PB[\delta, \lambda]$.

$$\begin{aligned} PB[\delta, \lambda] : \quad & \max \delta E(w_T) - \lambda E(w_T)^2 \\ \text{s.t. } & w_T = h_T^0 + \sum_{i=1}^N h_T^i \\ & h_{t+1}^0 = r^0 \left(h_t^0 - \sum_{i=1}^N (1 + \alpha^i) x_t^{i+} + \sum_{i=1}^N (1 - \beta^i) x_t^{i-} \right) \\ & h_{t+1}^i = R_t^i \left(h_t^i + x_t^{i+} - x_t^{i-} \right) \\ & x_t^{i+}, x_t^{i-} \geq 0. \end{aligned} \quad (8)$$

for $i = 1, 2, \dots, N$, and $t = 0, 1, \dots, T - 1$. For notational convenience, we use $E(\cdot)^2$ to represent $E[(\cdot)^2]$ and $E^2(\cdot)$ to represent $[E(\cdot)]^2$ throughout this paper.

3 Analytical solution to a two-period investment problem

To obtain some insights on the multiperiod problem, we start with studying a two-period problem with a market consisting of two assets, one risk-free asset such as a cash deposit and one risky asset such as a mutual fund, denoted by asset 0 and asset 1 respectively. The properties of the optimal solutions are given in Proposition 1.

Proposition 1 *Under the optimal investment policy, at least one of the two decision variables at each stage should equal zero.*

Proof Let U be the objective function in $PB[\lambda]$, and $\mathbf{x}_t = (x_t^{1+}, x_t^{1-})$, for $t = 0, 1, \dots, T - 1$. Assume that there exists a feasible investment policy for $PB[\lambda]$ that is represented by $\hat{\pi} =$

$(\dot{x}_0, \dots, \dot{x}_s, \dots, \dot{x}_{T-1})$, where $\dot{x}_s$ contains two positive elements $\dot{x}_s^{1+}$ and $\dot{x}_s^{1-}$. Suppose there is another feasible policy $\tilde{\pi}$ which is identical to π in all the periods except period s where

$$\begin{cases} \ddot{x}_s^{1+} = \dot{x}_s^{1+} - \dot{x}_s^{1-} \\ \ddot{x}_s^{1-} = 0 \end{cases} \quad \text{if } \dot{x}_s^{1+} \geq \dot{x}_s^{1-},$$

or

$$\begin{cases} \ddot{x}_s^{1+} = 0 \\ \ddot{x}_s^{1-} = \dot{x}_s^{1-} - \dot{x}_s^{1+} \end{cases} \quad \text{if } \dot{x}_s^{1+} \leq \dot{x}_s^{1-}.$$

The two policies make the same adjustment on the risky asset while $\tilde{\pi}$ incurs more transaction costs that are withdrawn from the risk-free account.

If $\dot{x}_s^{1+} \geq \dot{x}_s^{1-}$, then

$$U|_{\tilde{\pi}} = U|_{\pi} - (r^0)^{T-s}(\alpha + \beta)\dot{x}_s^{1-} < U|_{\pi}.$$

This means that a policy can always be improved for $PB[\lambda]$ if there are both purchasing and selling actions at the same stage. In other words, a necessary condition for an optimal solution is

$$x_t^{1+} \cdot x_t^{1-} = 0 \quad \text{for } t = 0, 1, \dots, T-1.$$

Similarly, we have the same conclusion for the case of $\dot{x}_s^{1+} \leq \dot{x}_s^{1-}$. As $PB[\lambda]$ and $PB[\delta, \lambda]$ have the same optimal solution, this property also applies to the latter. $\square$

3.1 Analytical solution to the second period

Assume that the solution to the first period has been obtained. For the second period, the value function that can be used in a dynamic programming approach is

$$V_1(x_1^{1+}, x_1^{1-} | h_1^0, h_1^1) = \delta E(h_2^0 + h_2^1) - \lambda E(h_2^0 + h_2^1)^2. \quad (9)$$

The one-period optimization problem is then modeled as follows:

$$\begin{aligned} \max \quad & V_1(x_1^{1+}, x_1^{1-} | h_1^0, h_1^1) \\ \text{s.t.} \quad & h_2^0 = r^0(h_1^0 - (1 + \alpha)x_1^{1+} + (1 - \beta)x_1^{1-}) \\ & h_2^1 = R_1^1(h_1^1 + x_1^{1+} - x_1^{1-}) \\ & x_1^{1+}, x_1^{1-} \geq 0. \end{aligned} \quad (10)$$

Define

$$\xi_1^\alpha = R_1^1 - r^0(1 + \alpha) \quad (11)$$

$$\xi_1^\beta = R_1^1 - r^0(1 - \beta) \quad (12)$$

$$K_1^\alpha = \frac{E^2(\xi_1^\alpha)}{E(\xi_1^\alpha)^2} \quad (13)$$

$$K_1^\beta = \frac{E^2(\xi_1^\beta)}{E(\xi_1^\beta)^2} \quad (14)$$

$$J_1^\alpha = \frac{E(R_1^1 \xi_1^\alpha)}{E(\xi_1^\alpha)} \quad (15)$$

Table 1 Solutions to the last stage of PB[δ, λ]

Range of $\frac{\delta}{2\lambda}$	Solution and value function $V_1(h_1^0, h_1^1)$
$[G_1^\alpha, +\infty)$	$x_1^{1+} = \frac{E(\xi_1^\alpha)}{E(\xi_1^\alpha)^2} \left(\frac{\delta}{2\lambda} - G_1^\alpha \right), \quad x_1^{1-} = 0$ $V_1^{(1)} = \delta E \left(r^0 h_1^0 + R_1^1 h_1^1 \right) - \lambda E \left(r^0 h_1^0 + R_1^1 h_1^1 \right)^2 + K_1^\alpha \lambda \left(\frac{\delta}{2\lambda} - G_1^\alpha \right)^2$
$[G_1^\beta, G_1^\alpha)$	$x_1^{1+} = 0, \quad x_1^{1-} = 0$ $V_1^{(2)} = \delta E \left(r^0 h_1^0 + R_1^1 h_1^1 \right) - \lambda E \left(r^0 h_1^0 + R_1^1 h_1^1 \right)^2$
$[0, G_1^\beta)$	$x_1^{1+} = 0, \quad x_1^{1-} = \frac{E(\xi_1^\beta)}{E(\xi_1^\beta)^2} \left(G_1^\beta - \frac{\delta}{2\lambda} \right)$ $V_1^{(3)} = \delta E \left(r^0 h_1^0 + R_1^1 h_1^1 \right) - \lambda E \left(r^0 h_1^0 + R_1^1 h_1^1 \right)^2 + K_1^\beta \lambda \left(\frac{\delta}{2\lambda} - G_1^\beta \right)^2$

$$J_1^\beta = \frac{E(R_1^1 \xi_1^\beta)}{E(\xi_1^\beta)} \quad (16)$$

$$G_1^\alpha = r^0 h_1^0 + J_1^\alpha h_1^1 \quad (17)$$

$$G_1^\beta = r^0 h_1^0 + J_1^\beta h_1^1. \quad (18)$$

ξ_1^α and ξ_1^β can be interpreted as the excess return. We assume $E(R_1^1) \geq r^0(1+\alpha)$ to ensure that the investor expects a higher return by taking up extra risk. It can be easily verified that $E(\xi_1^\alpha) < E(\xi_1^\beta)$, $K_1^\alpha < K_1^\beta$, $J_1^\alpha > J_1^\beta$ and $G_1^\alpha > G_1^\beta$, for $h_1^1 \geq 0$.

The results of solving the model are summarized in Table 1. It can be seen that G_1^α and G_1^β are two thresholds for change in investment strategy. They divide the action space into three regions. According to the range that $\frac{\delta}{2\lambda}$ falls into, three investment strategies are recommended respectively.

In the first interval, when $G_1^\alpha < \frac{\delta}{2\lambda}$, the sum of the expected return from the current portfolio and the risk level of purchasing more risky asset is smaller than $\frac{\delta}{2\lambda}$. Hence, the corresponding optimal policy is to buy the risky asset. On the other hand, for the third interval, the sum of the expected return and the risk level of keeping all the risky asset exceeds the limit of $\frac{\delta}{2\lambda}$. Therefore, the suggested policy is to reduce the holding of the risky asset.

Lastly, between the first and third intervals, there exists a non-transaction region when $G_1^\beta \leq \frac{\delta}{2\lambda} \leq G_1^\alpha$. The investor is recommended to take no action since the increase of the investor's expected utility cannot compensate for the transaction costs incurred from rebalancing the portfolio.

The optimal value function that we obtained is found to be smooth on the entire domain. We also compare our solution with the closed-form solution obtained by Li and Ng (2000) for the multiperiod mean-variance problem *without* transaction costs. In particular, if we let the rates of transaction cost, α and β , take the value of zero, the non-transaction region disappears and the optimal investment policy will be to rebalance asset levels to

$$h_1^0 - x_1^{1+} + x_1^{1-} = -\frac{E(\xi_1)}{E(\xi_1)^2} \frac{\delta}{2\lambda} + \frac{E(R_1^1 \xi_1)}{E(\xi_1)^2} (h_1^0 + h_1^1), \quad (19)$$

where $\xi_1 = R_1^1 - r^0$, which agrees with Li and Ng's results.

3.2 Analytical solution to the first period

At the beginning of stage 1, we are faced with a two-period problem ahead (stage 1 and stage 2). Solving the two-stage problem is the same as finding the optimal actions to take in the next two time periods, i.e., looking for the optimal values of x_0^{1+} , x_0^{1-} , x_1^{1+} and x_1^{1-} . Given the initial asset level of h_0^0 and h_0^1 at the beginning of stage 1, we have to decide how much risky asset to buy or sell (x_0^{1+} and x_0^{1-}), and the action taken will affect the asset levels of the next stage, h_1^0 and h_1^1 . Once h_1^0 and h_1^1 are known, we can find the best actions at stage 2 from Table 1 and the final reward is immediately known. Thus, we can obtain the optimal set of actions to maximize the final reward.

We apply backward dynamic programming for solving the problem. For the first stage, we introduce a new set of post-decision state variables $\hat{h}_t^0$ and $\hat{h}_t^1$ to represent the asset levels immediately after the transactions have been performed:

$$\hat{h}_t^0 = h_t^0 - (1 + \alpha)x_t^{1+} + (1 - \beta)x_t^{1-} \quad (20)$$

$$\hat{h}_t^1 = h_t^1 + x_t^{1+} - x_t^{1-}. \quad (21)$$

Combining (2), (3), (20) and (21) yields

$$h_{t+1}^0 = r^0 \hat{h}_t^0 \quad (22)$$

$$h_{t+1}^1 = R_t^1 \hat{h}_t^1. \quad (23)$$

Thus, the two thresholds G_1^α and G_1^β can be rewritten as

$$G_1^\alpha = (r^0)^2 \hat{h}_0^0 + J_1^\alpha R_0^1 \hat{h}_0^1 \quad (24)$$

$$G_1^\beta = (r^0)^2 \hat{h}_0^0 + J_1^\beta R_0^1 \hat{h}_0^1. \quad (25)$$

From equations (24) and (25), we can derive the two thresholds for R_0^1 :

$$B_0^\alpha = \frac{\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0}{J_1^\alpha \hat{h}_0^1} \quad (26)$$

$$B_0^\beta = \frac{\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0}{J_1^\beta \hat{h}_0^1}. \quad (27)$$

Then the value function for the first stage can be written as

$$V_0 = \max_a \int_a^{B_0^\alpha} p(r_0^1) V_1^{(1)} dr_0^1 + \int_{B_0^\alpha}^{B_0^\beta} p(r_0^1) V_1^{(2)} dr_0^1 + \int_{B_0^\beta}^b p(r_0^1) V_1^{(3)} dr_0^1. \quad (28)$$

In the multiperiod setting, considering sophisticated distributions adds much difficulty for solving. Therefore, existing studies often choose simple return rate assumptions, e.g., Gennotte and Jung (1994) assume a distribution with two possible outcomes, Bertsimas and Pachamanova (2008) use worst-case returns with correlations for robust optimization. Instead of considering only a finite set of realizations of the return rate, we use the uniform distribution assumption for the first period in our paper for its simplicity and frequent usage in financial analysis (McFarland 1988; Shalit and Yitzhaki 2002; Kuan et al. 2009; Wagner 2010). The only information needed to apply a uniform distribution is the upper bound and lower bound of the return rate. With the uniform distribution assumption, a closed-form solution to the

two-period problem can be obtained. It is also worth noting that our analysis for the second period is not affected by the choice of probability distribution of the return rate for the first period, i.e., there will always exist three solution intervals as long as the transaction costs are present.

We assume $R_0^1 \sim U(a, b)$, where a and b satisfy $b > \frac{r^0(1+\alpha)}{1-\beta}$ and $a < r^0$ to ensure that investing in one asset is not strictly dominated by the other. Additionally, all the wealth is assumed to be deposited in the risk-free account at the beginning of the first period, i.e., $h_0^0 = w_0$ and $h_0^1 = 0$. The investor decides the amount of wealth to be shifted from the risk-free asset to the risky asset. Therefore, $x_0^{1+} \geq 0$ and $x_0^{1-} = 0$. The resulting value function for the first period is

$$V_0 = \delta E \left[(r^0)^2 \hat{h}_0^0 + R_1^1 R_0^1 \hat{h}_0^1 \right] - \lambda E \left[(r^0)^2 \hat{h}_0^0 + R_1^1 R_0^1 (\hat{h}_0^1) \right]^2 - \frac{\lambda}{3(b-a)\hat{h}_0^1} \\ \times \left\{ \frac{K_1^\beta}{J_1^\beta} \left[\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\beta b \hat{h}_0^1 \right]^3 - \frac{K_1^\alpha}{J_1^\alpha} \left[\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha a \hat{h}_0^1 \right]^3 \right\}. \quad (29)$$

The derivation details can be found in Appendix 1.

In order to find the global optima to the problem, the concavity of V_0 needs to be established. As such, we further define a random variable

$$\xi_0^\alpha = R_1^1 R_0^1 - (r^0)^2 (1 + \alpha) \quad (30)$$

and two constants

$$L_0^b = J_1^\beta b - (r^0)^2 (1 + \alpha) \quad (31)$$

$$L_0^a = J_1^\alpha a - (r^0)^2 (1 + \alpha). \quad (32)$$

Then we have the following results:

Lemma 1 $E(\xi_0^\alpha)^2 > \frac{\frac{K_1^\beta}{J_1^\beta} (L_0^b)^3 - \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^3}{3(b-a)}.$

Proof

$$E(\xi_0^\alpha)^2 = \frac{\frac{K_1^\beta}{J_1^\beta} (L_0^b)^3 - \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^3}{3(b-a)} \\ = E(R_1^1)^2 \frac{b^3 - a^3}{3(b-a)} - E(R_1^1)(a+b)(r^0)^2(1+\alpha) + \left[(r^0)^2(1+\alpha) \right]^2 \\ - \frac{1}{3(b-a)} \left[\left(K_1^\beta (J_1^\beta)^2 b^3 - K_1^\alpha (J_1^\alpha)^2 a^3 \right) - 3 \left(K_1^\beta J_1^\beta b^2 - K_1^\alpha J_1^\alpha a^2 \right) (r^0)^2(1+\alpha) \right. \\ \left. + 3 \left(K_1^\beta b - K_1^\alpha a \right) \left((r^0)^2(1+\alpha) \right)^2 - \left(\frac{K_1^\beta}{J_1^\beta} - \frac{K_1^\alpha}{J_1^\alpha} \right) \left((r^0)^2(1+\alpha) \right)^3 \right] \\ = \frac{\text{Var}(R_1^1)}{3(b-a)r^0} \left[\frac{\left(r^0(1-\beta)b - (r^0)^2(1+\alpha) \right)^3}{E(\xi_1^\beta)(1-\beta)} - \frac{(a-r^0)^3}{E(\xi_1^\alpha)(1+\alpha)} \right]$$

$$\begin{aligned}
& + \frac{\text{Var}(R_1^1)}{r^0(1+\alpha)(1-\beta)E(R_1^1\xi_1^\alpha)E(R_1^1\xi_1^\beta)} \left[\left(\alpha E(\xi_1^\alpha) + \beta E(\xi_1^\beta) \right) E(R_1^1 - r^0) \right. \\
& \left. + (\alpha + \beta) \text{Var}(R_1^1) + r^0(\alpha + \beta) E(R_1^1 - r^0(1 + \alpha\beta)) \right] > 0.
\end{aligned}$$

□

Lemma 2 $\frac{\delta}{2\lambda} > (r^0)^2 \hat{h}_0^0 + (r^0)^2(1 + \alpha)\hat{h}_0^1$ under the optimal investment policy π^* .

Proof As $\text{PB}(\lambda)$ is equivalent to $\text{PB}(\delta, \lambda)$ if and only if $\delta = 1 + 2\lambda E(w_T)|_{\pi^*}$, therefore under the optimal investment policy,

$$\begin{aligned}
\frac{\delta}{2\lambda} &= \frac{1}{2\lambda} + E(w_T)|_{\pi^*} \\
&\geq \frac{1}{2\lambda} + (r^0)^2 \hat{h}_0^0 + E(R_1^1)E(R_1^1)\hat{h}_0^1 \\
&> (r^0)^2 \hat{h}_0^0 + (r^0)^2(1 + \alpha)\hat{h}_0^1.
\end{aligned} \tag{33}$$

□

Lemmas 1 and 2 lay a preliminary foundation for proving the concavity of V_0 . It will be shown in Theorem 1 that V_0 is strictly concave with respect to x_0^{1+} and the KKT conditions can be applied for solving the problem.

Theorem 1 The KKT conditions are sufficient for finding the optimal solution to the two-period problem that has the objective function given by (29).

Proof From Lemmas 1 and 2, we can derive

$$\begin{aligned}
\frac{d^2 V_0(x_0^{1+})}{d(x_0^{1+})^2} &= -2\lambda \left[E(\xi_0^\alpha)^2 - \frac{\frac{K_1^\beta}{J_1^\beta} (L_0^b)^3 - \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^3}{3(b-a)} \right] \\
&\quad - \left(\frac{K_1^\beta}{J_1^\beta} - \frac{K_1^\alpha}{J_1^\alpha} \right) \left(\frac{\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - (r^0)^2(1 + \alpha)\hat{h}_0^1}{\hat{h}_0^1} \right)^3 < 0
\end{aligned}$$

Hence, the objective function is a concave function of x_0^{1+} , and the KKT conditions are sufficient for finding the optimal solution. □

Theorem 1 allows the optimal solution to be obtained by solving the following KKT conditions:

- (i) $\frac{dV_0(x_0^{1+})}{dx_0^{1+}} - y = 0$
- (ii) $x_0^{1+} \geq 0$
- (iii) $y \geq 0$
- (iv) $x_0^{1+}y = 0$

where y is the Lagrange multiplier.

Corollary 1 *The optimal solution for the first period is*

$$x_0^{1+*} = \frac{\frac{\gamma}{2} - (r^0)^2 w_0}{u_0} \quad (34)$$

$$x_0^{1-*} = 0 \quad (35)$$

where $\gamma = \delta/\lambda$.

Proof Let $y = 0$. Then condition (i) becomes

$$\begin{aligned} 0 &= \frac{dV_0(x_0^{1+})}{dx_0^{1+}} \\ &= \delta E(\xi_0^\alpha) - 2\lambda E\left[\xi_0^\alpha \left((r^0)^2 h_0^0 + R_1^1 R_0^1 h_0^1 + \xi_0^\alpha x_0^{1+}\right)\right] \\ &\quad + \frac{\lambda}{3(b-a)(x_0^{1+})^2} \left\{ 3 \frac{K_1^\beta}{J_1^\beta} \left[\frac{\delta}{2\lambda} - (r^0)^2 w_0 - L_0^b x_0^{1+} \right]^2 L_0^b x_0^{1+} \right. \\ &\quad - 3 \frac{K_1^\alpha}{J_1^\alpha} \left[\frac{\delta}{2\lambda} - (r^0)^2 h_0^0 - L_0^a x_0^{1+} \right]^2 L_0^a x_0^{1+} + \frac{K_1^\beta}{J_1^\beta} \left[\frac{\delta}{2\lambda} - (r^0)^2 w_0 - L_0^b x_0^{1+} \right]^3 \\ &\quad \left. - \frac{K_1^\alpha}{J_1^\alpha} \left[\frac{\delta}{2\lambda} - (r^0)^2 w_0 - L_0^a x_0^{1+} \right]^3 \right\}. \end{aligned} \quad (36)$$

Multiplying both sides by $\frac{3(b-a)}{\lambda x_0^{1+}}$ yields

$$\begin{aligned} 0 &= \left(\frac{K_1^\beta}{J_1^\beta} - \frac{K_1^\alpha}{J_1^\alpha} \right) \left(\frac{\frac{\delta}{2\lambda} - (r^0)^2 w_0}{x_0^{1+}} \right)^3 \\ &\quad + 3 \left(2(b-a) E(\xi_0^\alpha) - \frac{K_1^\beta}{J_1^\beta} (L_0^b)^2 + \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^2 \right) \left(\frac{\frac{\delta}{2\lambda} - (r^0)^2 w_0}{x_0^{1+}} \right) \\ &\quad - 6(b-a) E(\xi_0^\alpha)^2 + 2 \frac{K_1^\beta}{J_1^\beta} (L_0^b)^3 - 2 \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^3. \end{aligned} \quad (37)$$

Let

$$u = \frac{\frac{\delta}{2\lambda} - (r^0)^2 w_0}{x_0^{1+}} \quad (38)$$

$$A = \frac{K_1^\beta}{J_1^\beta} - \frac{K_1^\alpha}{J_1^\alpha} \quad (39)$$

$$B = 3 \left(2(b-a) E(\xi_0^\alpha) - \frac{K_1^\beta}{J_1^\beta} (L_0^b)^2 + \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^2 \right) \quad (40)$$

$$C = -6(b-a) E(\xi_0^\alpha)^2 + 2 \frac{K_1^\beta}{J_1^\beta} (L_0^b)^3 - 2 \frac{K_1^\alpha}{J_1^\alpha} (L_0^a)^3. \quad (41)$$

Then (37) can be written as

$$Au^3 + Bu + C = 0 \quad (42)$$

As proven in Lemma 1, $\frac{C}{A}$ is negative, which implies that (42) has at least one positive root. Additionally, V_0 is strictly concave and so there can at most be one optimal solution. Denoting u_0 as the positive root of (42), x_0^{1+*} can be expressed as

$$x_0^{1+*} = \frac{\frac{\gamma}{2} - (r^0)^2 w_0}{u_0}, \quad (43)$$

where $\gamma = \delta/\lambda$. $\square$

With the optimal policy for the first period provided in Corollary 1, we can further derive the mean-variance frontier of our model. The derivation details are as follows:

The targeted holdings of assets under such a policy are given by

$$\begin{aligned} \hat{h}_0^0 &= \frac{-(1+\alpha)\frac{\gamma}{2} + J_0^\alpha w_0}{J_0^\alpha - (r^0)^2(1+\alpha)} \\ \hat{h}_0^1 &= \frac{\frac{\gamma}{2} - (r^0)^2 w_0}{J_0^\alpha - (r^0)^2(1+\alpha)}, \end{aligned}$$

where $J_0^\alpha = u_0 + (r^0)^2(1+\alpha)$. The expected final wealth is (see Appendix 2 for derivation details)

$$\begin{aligned} E(w_2(\gamma)) &= \int_a^{B_{T-2}^\alpha} p(r_0^1) w_1^{(1)} dr_0^1 + \int_{B_{T-2}^\beta}^{B_{T-2}^\alpha} p(r_0^1) w_1^{(2)} dr_0^1 + \int_{B_{T-2}^\beta}^b p(r_0^1) w_1^{(3)} dr_0^1 \\ &= (r^0)^2 w_0 + \rho \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{J_0^\alpha - (r^0)^2(1+\alpha)} \right), \end{aligned} \quad (44)$$

where

$$\rho = E(\xi_0^\alpha) + \frac{1}{2(b-a)} \frac{K_1^\alpha}{J_1^\alpha} (J_0^\alpha - J_1^\alpha a)^2 - \frac{1}{2(b-a)} \frac{K_1^\beta}{J_1^\beta} (J_0^\alpha - J_1^\beta b)^2. \quad (45)$$

Furthermore, we have

$$\begin{aligned} E(w_2(\gamma))^2 &= \int_a^{B_{T-2}^\alpha} p(r_0^1) (w_1^{(1)})^2 dr_0^1 + \int_{B_{T-2}^\beta}^{B_{T-2}^\alpha} p(r_0^1) (w_1^{(2)})^2 dr_0^1 + \int_{B_{T-2}^\beta}^b p(r_0^1) (w_1^{(3)})^2 dr_0^1 \\ &= v \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{J_0^\alpha - (r^0)^2(1+\alpha)} \right)^2 + 2\rho(r^0)^2 w_0 \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{J_0^\alpha - (r^0)^2(1+\alpha)} \right) + ((r^0)^2 w_0)^2, \end{aligned} \quad (46)$$

where

$$\begin{aligned} v &= E(\xi_0^\alpha)^2 + \frac{1}{3(b-a)} \left[\frac{K_1^\beta}{J_1^\beta} (J_0^\alpha - J_1^\beta b)^3 - \frac{K_1^\alpha}{J_1^\alpha} (J_0^\alpha - J_1^\alpha a)^3 \right] \\ &\quad - \frac{1}{b-a} \left[\frac{K_1^\beta}{J_1^\beta} (J_0^\alpha - J_1^\beta b)^2 - \frac{K_1^\alpha}{J_1^\alpha} (J_0^\alpha - J_1^\alpha a)^2 \right]. \end{aligned} \quad (47)$$

Combining (44) and (46) yields

$$\text{Var}(w_2(\gamma)) = E(w_2(\gamma))^2 - E^2(w_2(\gamma)) = (v - \rho^2) \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{u_0^*} \right)^2. \quad (48)$$

Thus, letting $U = E(w_2(\gamma)) - \lambda \text{Var}(w_2(\gamma))$, we have

$$U = (r^0)^2 w_0 + \rho \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{u_0^*} \right) - \lambda (v - \rho^2) \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{u_0^*} \right)^2. \quad (49)$$

Differentiating (49) with respect to γ gives

$$\frac{dU}{d\gamma} = \frac{\rho}{u_0^*} - 2\lambda \frac{v - \rho^2}{u_0^*} \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{u_0^*} \right). \quad (50)$$

Solving $\frac{\partial \tilde{U}}{\partial \gamma} = 0$ yields

$$\gamma^* = \frac{\rho u_0^*}{\lambda(v - \rho^2)} + 2(r^0)^2 w_0. \quad (51)$$

Substituting (51) into (44) and (48), we have

$$E(w_2(\gamma^*)) = (r^0)^2 w_0 + \frac{\rho^2}{2\lambda(v - \rho^2)}, \quad (52)$$

$$\text{Var}(w_2(\gamma^*)) = \frac{\rho^2}{4\lambda^2(v - \rho^2)}. \quad (53)$$

The mean-variance efficient frontier is then

$$\text{Var}(w_2) = \frac{v - \rho^2}{\rho^2} \left(E(w_2) - (r^0)^2 w_0 \right)^2. \quad (54)$$

$\text{Var}(w_2)$ is a quadratic function of $E(w_2)$ in (54), indicating a risk-averse utility function. Additionally, when $E(w_2)$ is equal to $(r^0)^2 w_0$, $\text{Var}(w_2) = 0$. It means that if all the wealth has been allocated to the risk-free asset in the first and second periods, there is no uncertainty in this investment strategy.

4 Numerical example

Suppose an investor with 10 million dollars is to make a yearly investment plan with a rebalancing time point at the end of the sixth month. The allocation is based on one risky asset and one risk-free asset. The risk-free asset has a return rate of 1.04, while the return rate of the risky asset follows a uniform distribution $U(0.95, 1.3)$ for both periods. λ takes the value of 0.5 to represent the investor's risk profile. The buying and selling transaction costs are both 1.5 %. The optimal strategy for the first period is then

$$x_0^{1+*} = 9.4849 \quad \text{and} \quad x_0^{1-*} = 0, \quad (55)$$

while at the second stage, the optimal strategy is

$$\begin{cases} x_1^{1+*} = 4.1478(12.4871 - 12.2989R_0^1), x_1^{1-*} = 0, & \text{if } 0.95 \leq R_0^1 < 1.0153 \\ x_1^{1+*} = 0, x_1^{1-*} = 0, & \text{if } 1.0153 \leq R_0^1 < 1.0735 \\ x_1^{1+*} = 0, x_1^{1-*} = 4.9500(11.6324R_0^1 - 12.4871), & \text{if } 1.0735 \leq R_0^1 \leq 1.3 \end{cases} \quad (56)$$

The optimal portfolio yields an expected final wealth of 11.8904 million dollars with a variance of 1.0744 square million dollars.

In (56), 1.0153 and 1.0735 are the two thresholds where the optimal action changes. When R_0^1 is below 1.0153, a relatively low level of holdings of the risky asset would occur at the end of the first period, and therefore the optimal solution is to buy more risky asset. On the other hand, if R_0^1 exceeds 1.0735, the investor should shift the wealth from the risky asset to the risk-free asset in order to reduce the risk. When R_0^1 is between 1.0153 and 1.0735, $x_1^{1+*} = x_1^{1-*} = 0$, which means that no adjustment should be made to the current portfolio. This is the no-transaction region.

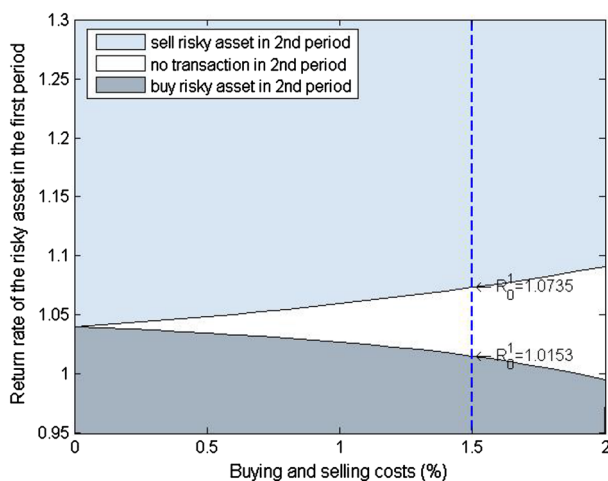


Fig. 1 Optimal investment strategy in the second period when $\alpha = \beta$

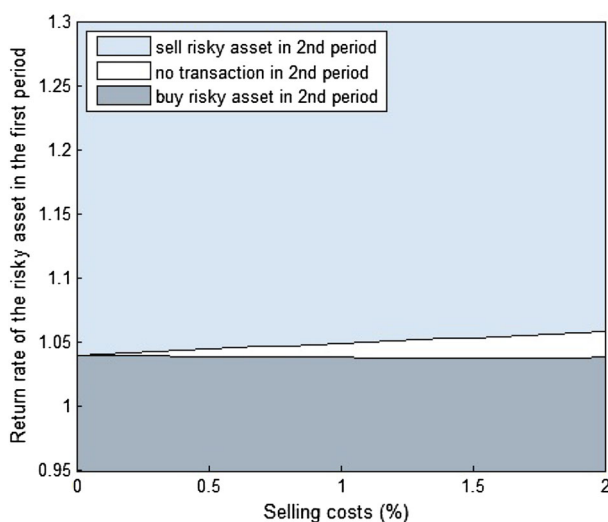


Fig. 2 Optimal investment strategy in the second period when $\alpha = 0$

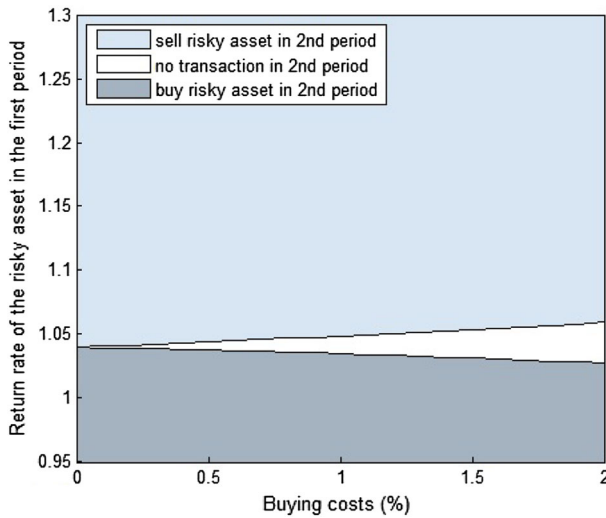


Fig. 3 Optimal investment strategy in the second period when $\beta = 0$

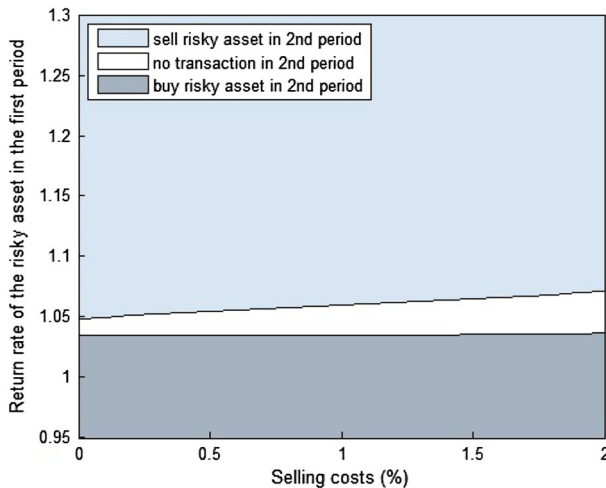


Fig. 4 Optimal investment strategy in the second period when $\alpha = 1\%$

In conclusion, at the end of the first period, the realization of R_0^1 can be observed. It will decide the asset levels at the beginning of the second period (h_1^0 and h_1^1). As shown in Table 1, based on the observation of h_1^0 and h_1^1 , the investor can easily find the best investment strategy for the second period. In Fig. 1, we plotted values of the two thresholds for R_0^1 by changing the value of transaction costs (α and β), while keeping other data the same as those in the example above. As shown in Fig. 1, when there is no transaction cost ($\alpha = \beta = 0$), the no-transaction region is empty, i.e., the optimal strategy is either buying or selling the asset. As the transaction rates increase, the no-transaction region enlarges. It means that the optimal strategy is to transact less frequently as the transaction becomes more expensive.

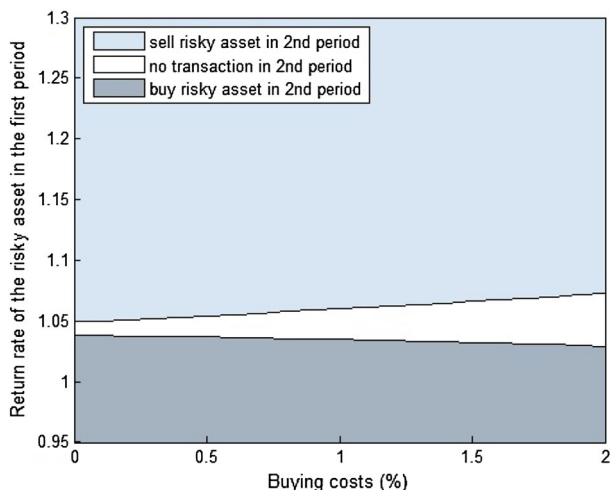


Fig. 5 Optimal investment strategy in the second period when $\beta = 1\%$

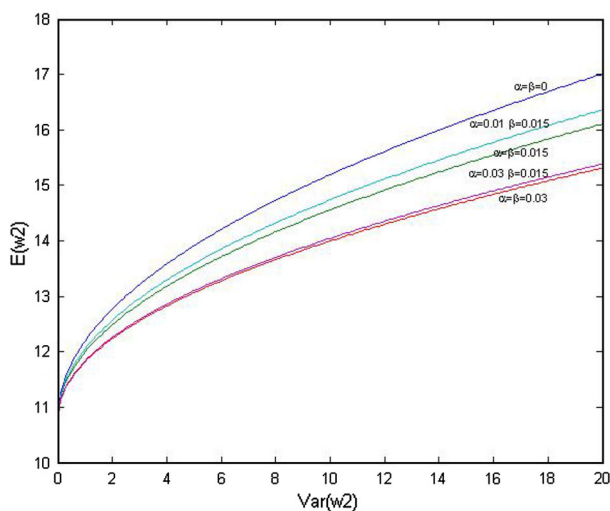


Fig. 6 Markowitz efficient frontiers with proportional transaction costs

Figures 2, 3, 4 and 5 plot the values of the two thresholds for R_0^1 for the cases of $\alpha = 0$, $\beta = 0$, $\alpha = 1\%$ and $\beta = 1\%$ respectively. In Figs. 2 and 3, there is transaction cost in either buying or selling only. As a result, the overall transaction costs are lower than that in Fig. 1, and thus the no-transaction regions are also smaller. The investor is encouraged to be more active in transacting when either the buying or selling fee is eliminated. In a two-period investment problem, when buying (selling) is less costly, the investor may sell (buy) more in the first period since the investor can buy (sell) with a low cost in the second period. Therefore, lowered fee for either buying or selling may lead to more transactions in buying and selling.

When α or β is a positive constant, such as the scenarios shown in Figs. 4 and 5, the no-transaction region is not empty. In Fig. 4, as the value of β increases, the no-transaction region enlarges gradually. In the left half of Fig. 4 where $\beta \leq 1\%$, the no-transaction region

is larger than that in the left half of Fig. 1. When $\beta > 1\%$, the no-transaction region in Fig. 4 is smaller than that in Fig. 1. This is because the overall transaction costs are higher in the left half of Fig. 4 than in Fig. 1, while it is the reverse situation for the right half of the figures. Similar observations can be obtained from Fig. 5.

Figure 6 shows the efficient frontiers using the data given above but under five transaction fee schedules. As the overall transaction costs increase, the efficient frontier moves downwards. The five frontiers meet at one end when $\text{Var}(w_2) = 0$ and $E(w_2) = 10.8160$. In this situation, the investor tends to avoid any risk. Therefore the optimal strategy is always to invest in the risk-free asset no matter how much the transaction fee is. As the transaction fee increases, the expectation of the investor's final wealth decreases for the same exposure to investment risk. If two efficient frontiers have the same buying (selling) costs, the efficient frontier with the lower selling (buying) costs will be higher than the other. The efficient frontier with no transaction costs ($\alpha = \beta = 0$) is an upper bound of efficient frontiers with transaction fees. If transaction costs are ignored, an investor will be faced with a lower reward than expected from a portfolio constructed according to the investor's risk tolerance or exposed to a higher risk for the investor's target return.

5 Conclusion

In this paper, we consider a two-period investment problem that incorporates proportional transaction costs. The closed-form solution for a two-period problem is obtained and its characteristics are identified. This paper can be considered as a first attempt to solve the multiperiod mean-variance portfolio optimization problem with transaction costs. Our results show that the transaction costs will significantly affect the optimal investment decisions. The two thresholds where the investment strategy changes have been analytically identified. Furthermore, even for a two-period problem under the simple assumption of uniformly distributed return rate, the quadratic nature of the value function is not preserved. Therefore, it is very challenging to solve multiperiod problems with transaction costs and approximation techniques are needed to tackle this problem. A possible future research topic could be to explore effective and efficient approximation methods to solve large-scale multiperiod mean-variance portfolio optimization problems in the presence of transaction costs.

Appendix 1: Value function for the first period

We apply the backward dynamic programming algorithm to obtain the value function. In particular, after we have obtained the piecewise value function for the latter period, we can derive the value function for the first period as follows:

$$\begin{aligned}
 V_0(\hat{h}_0^0, \hat{h}_0^1) = & \max_{\alpha} \int_a^{B_0^\alpha} \frac{1}{b-a} \left[\delta E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right) - \lambda E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right)^2 \right. \\
 & \left. + K_1^\alpha \lambda \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha r_0^1 \hat{h}_0^1 \right)^2 \right] dr_0^1 \\
 & + \int_{B_0^\alpha}^{B_0^\beta} \frac{1}{b-a} \left[\delta E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right) - \lambda E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right)^2 \right] dr_0^1
 \end{aligned}$$

$$\begin{aligned}
& + \int_{B_0^\beta}^b \frac{1}{b-a} \left[\delta E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right) - \lambda E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right)^2 \right. \\
& \quad \left. + K_1^\beta \lambda \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\beta r_0^1 \hat{h}_0^1 \right)^2 \right] dr_0^1 \\
& = \max \int_a^b \frac{1}{b-a} \left[\delta E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right) - \lambda E \left((r^0)^2 \hat{h}_0^0 + R_1^1 r_0^1 \hat{h}_0^1 \right)^2 \right] dr_0^1 \\
& \quad + \int_a^{B_0^\alpha} \frac{1}{b-a} K_1^\alpha \lambda \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha r_0^1 \hat{h}_0^1 \right)^2 dr_0^1 \\
& \quad + \int_{B_0^\beta}^b \frac{1}{b-a} K_1^\beta \lambda \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\beta r_0^1 \hat{h}_0^1 \right)^2 dr_0^1. \tag{57}
\end{aligned}$$

Since

$$\left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha r_0^1 \hat{h}_0^1 \right)^3 \Big|_{r_{T-2}^1 = B_{T-2}^\alpha} = 0 \tag{58}$$

$$\left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\beta r_0^1 \hat{h}_0^1 \right)^3 \Big|_{r_{T-2}^1 = B_{T-2}^\beta} = 0, \tag{59}$$

the value function can be simplified to

$$\begin{aligned}
V_0(\hat{h}_0^0, \hat{h}_0^1) & = \max - \frac{1}{b-a} \left\{ \delta \left[(r^0)^2 \hat{h}_0^0 a + \frac{1}{2} E(R_1^1) \hat{h}_0^1 a^2 \right] \right. \\
& \quad - \lambda \left[\left((r^0)^2 \hat{h}_0^0 \right)^2 a + (r^0)^2 E(R_1^1) \hat{h}_0^0 \hat{h}_0^1 a^2 + \frac{1}{3} E(R_1^1)^2 \left(\hat{h}_0^1 \right)^2 a^3 \right] \\
& \quad \left. - \frac{1}{3} K_1^\alpha \lambda \left(J_1^\alpha \hat{h}_0^1 \right)^{-1} \left[\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha \hat{h}_0^1 a \right]^3 \right\} \\
& \quad + \frac{1}{b-a} \left\{ \delta \left[(r^0)^2 \hat{h}_0^0 b + \frac{1}{2} E(R_1^1) \hat{h}_0^1 b^2 \right] \right. \\
& \quad - \lambda \left[\left((r^0)^2 \hat{h}_0^0 \right)^2 b + (r^0)^2 E(R_1^1) \hat{h}_0^0 \hat{h}_0^1 b^2 + \frac{1}{3} E(R_1^1)^2 \left(\hat{h}_0^1 \right)^2 b^3 \right] \\
& \quad \left. - \frac{1}{3} K_1^\beta \lambda \left(J_1^\beta \hat{h}_0^1 \right)^{-1} \left[\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\beta \hat{h}_0^1 b \right]^3 \right\} \tag{60} \\
& = \max \delta \left[(r^0)^2 \hat{h}_0^0 + E(R_1^1) \hat{h}_0^1 \frac{a+b}{2} \right] - \lambda \left[\left((r^0)^2 \hat{h}_0^0 \right)^2 \right. \\
& \quad \left. + (r^0)^2 E(R_1^1) \hat{h}_0^0 \hat{h}_0^1 (a+b) + \frac{1}{3} E(R_1^1)^2 \left(\hat{h}_0^1 \right)^2 (a^2 + ab + b^2) \right]
\end{aligned}$$

$$\begin{aligned}
& -\frac{\lambda}{b-a} \left\{ \frac{1}{3} \frac{\left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 \right)^3}{\hat{h}_0^1} \left(\frac{K_1^\beta}{J_1^\beta} - \frac{K_1^\alpha}{J_1^\alpha} \right) \right. \\
& \quad - \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 \right)^2 \left(K_1^\beta b - K_1^\alpha a \right) \\
& \quad + \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 \right) \hat{h}_0^1 \left(K_1^\beta J_1^\beta b^2 - K_1^\alpha J_1^\alpha a^2 \right) \\
& \quad \left. - \frac{1}{3} \left(\hat{h}_0^1 \right)^2 \left(K_1^\beta (J_1^\beta)^2 b^3 - K_1^\beta (J_1^\alpha)^2 a^3 \right) \right\}. \quad (61)
\end{aligned}$$

We assume that $R_0^1 \sim U(a, b)$, which implies $E(R_0^1) = \frac{a+b}{2}$ and $E(R_0^1)^2 = \frac{a^2+ab+b^2}{3}$. Therefore, the value function can be further simplified to

$$\begin{aligned}
V_0(\hat{h}_0^0, \hat{h}_0^1) &= \max \delta E \left[(r^0)^2 \hat{h}_0^0 + R_1^1 R_0^1 \hat{h}_0^1 \right] - \lambda E \left[(r^0)^2 \hat{h}_0^0 + R_{T-1}^1 R_0^1 (\hat{h}_0^1) \right]^2 \\
& - \frac{\lambda}{b-a} \left\{ \frac{1}{3} \frac{\left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 \right)^3}{\hat{h}_0^1} \left(\frac{K_1^\beta}{J_1^\beta} - \frac{K_1^\alpha}{J_1^\alpha} \right) - \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 \right)^2 \right. \\
& \quad \times \left(K_1^\beta b - K_1^\alpha a \right) + \left(\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 \right) \hat{h}_0^1 \left(K_1^\beta J_1^\beta b^2 - K_1^\alpha J_1^\alpha a^2 \right) \\
& \quad \left. - \frac{1}{3} \left(\hat{h}_0^1 \right)^2 \left(K_1^\beta (J_1^\beta)^2 b^3 - K_1^\beta (J_1^\alpha)^2 a^3 \right) \right\} \quad (62)
\end{aligned}$$

$$\begin{aligned}
&= \max \delta E \left[(r^0)^2 \hat{h}_0^0 + R_{T-1}^1 R_0^1 \hat{h}_0^1 \right] - \lambda E \left[(r^0)^2 \hat{h}_0^0 + R_{T-1}^1 R_0^1 (\hat{h}_0^1) \right]^2 \\
& - \frac{\lambda}{3(b-a)\hat{h}_0^1} \left\{ \frac{K_1^\beta}{J_1^\beta} \left[\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\beta b \hat{h}_0^1 \right]^3 \right. \\
& \quad \left. - \frac{K_1^\alpha}{J_1^\alpha} \left[\frac{\delta}{2\lambda} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha a \hat{h}_0^1 \right]^3 \right\}. \quad (63)
\end{aligned}$$

Appendix 2: Expressions for $E(w_2(\gamma))$ and $E(w_2(\gamma))^2$

The expectation of w_2 is

$$\begin{aligned}
E(w_2(\gamma)) &= \int_a^{B_{T-2}^\alpha} p(r_0^1) w_1^{(1)} dr_0^1 + \int_{B_{T-2}^\alpha}^{B_{T-2}^\beta} p(r_0^1) w_1^{(2)} dr_0^1 + \int_{B_{T-2}^\beta}^b p(r_0^1) w_1^{(3)} dr_0^1 \\
&= (r^0)^2 w_0 + \rho \left(\frac{\frac{\gamma}{2} - (r^0)^2 \hat{h}_0^0}{J_0^\alpha - (r^0)^2 (1 + \alpha)} \right), \quad (64)
\end{aligned}$$

where $\rho = E(\xi_0^\alpha) + \frac{1}{2(b-a)} \frac{K_1^\alpha}{J_1^\alpha} (J_0^\alpha - J_1^\alpha a)^2 - \frac{1}{2(b-a)} \frac{K_1^\beta}{J_1^\beta} (J_0^\alpha - J_1^\beta b)^2$.

The expectation of $(w_2)^2$ is

$$\begin{aligned}
 & E(w_2(\xi))^2 \\
 &= \int_a^{B_{T-2}^\alpha} p(r_0^1)(w_1^{(1)})^2 dr_0^1 + \int_{B_{T-2}^\alpha}^{B_{T-2}^\beta} p(r_0^1)(w_1^{(2)})^2 dr_0^1 + \int_{B_{T-2}^\beta}^b p(r_0^1)(w_1^{(3)})^2 dr_0^1 \\
 &= E \left[(r^0)^2 \hat{h}_0^0 + R_1^1 R_0^1 \hat{h}_0^1 \right]^2 + \frac{1}{(b-a)\hat{h}_0^1} \left\{ -\frac{1}{3} \frac{K_1^\alpha}{J_1^\alpha} \left(\frac{\gamma}{2} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha a \hat{h}_0^1 \right)^3 \right. \\
 &\quad + \frac{1}{3} \frac{K_1^\beta}{J_1^\beta} \left(\frac{\gamma}{2} - (r^0)^2 \hat{h}_0^0 - J_1^\beta b \hat{h}_0^1 \right)^3 \\
 &\quad + \frac{K_1^\alpha}{J_1^\alpha} \left(\frac{\gamma}{2} - (r^0)^2 \hat{h}_0^0 - J_1^\alpha a \hat{h}_0^1 \right)^2 \frac{\gamma}{2} \\
 &\quad \left. - \frac{K_1^\beta}{J_1^\beta} \left(\frac{\gamma}{2} - (r^0)^2 \hat{h}_0^0 - J_1^\beta b \hat{h}_0^1 \right)^2 \frac{\gamma}{2} \right\} \\
 &= v \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{J_0^\alpha - (r^0)^2 (1+\alpha)} \right)^2 + 2\rho (r^0)^2 w_0 \left(\frac{\frac{\gamma}{2} - (r^0)^2 w_0}{J_0^\alpha - (r^0)^2 (1+\alpha)} \right) + ((r^0)^2 w_0)^2,
 \end{aligned} \tag{65}$$

where

$$\begin{aligned}
 v = & E(\xi_0^\alpha)^2 + \frac{1}{3(b-a)} \left[\frac{K_1^\beta}{J_1^\beta} (J_0^\alpha - J_1^\beta b)^3 - \frac{K_1^\alpha}{J_1^\alpha} (J_0^\alpha - J_1^\alpha a)^3 \right] \\
 & - \frac{1}{b-a} \left[\frac{K_1^\beta}{J_1^\beta} (J_0^\alpha - J_1^\beta b)^2 - \frac{K_1^\alpha}{J_1^\alpha} (J_0^\alpha - J_1^\alpha a)^2 \right].
 \end{aligned} \tag{66}$$

References

- Bertsimas, D., & Pachamanova, D. (2008). Robust multiperiod portfolio management in the presence of transaction costs. *Computers and Operations Research*, 35(1), 3–17.
- Best, M., & Hlouskova, J. (2005). An algorithm for portfolio optimization with transaction costs. *Management Science*, 51(11), 1676–1688.
- Bi, J., Meng, Q., & Zhang, Y. (2014). Dynamic mean-variance and optimal reinsurance problems under the no-bankruptcy constraint for an insurer. *Annals of Operations Research*, 212(1), 43–59.
- Çelikyurt, U., & Özekici, S. (2007). Multiperiod portfolio optimization models in stochastic markets using the mean-variance approach. *European Journal of Operational Research*, 179(1), 186–202.
- Chen, A., Jen, F., & Zions, S. (1971). The optimal portfolio revision policy. *Journal of Business*, 44(1), 51–61.
- Chryssikou, E. (1998). *Multiperiod portfolio optimization in the presence of transaction costs*. Ph.D. thesis, Massachusetts Institute of Technology, Sloan School of Management.
- Constantinides, G. M. (1979). Multiperiod consumption and investment behavior with convex transactions costs. *Management Science*, 25(11), 1127–1137.
- Cox, J. C., & Huang, C.-F. (1989). Optimal consumption and portfolio policies when asset prices follow a diffusion process. *Journal of Economic Theory*, 49(1), 33–83.
- Davis, M., & Norman, A. (1990). Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4), 676–713.

- Dumas, B., & Luciano, E. (1991). An exact solution to a dynamic portfolio choice problem under transaction costs. *Journal of Finance*, 46(2), 577–595.
- Gennotte, G., & Jung, A. (1994). Investment strategies under transaction costs: the finite horizon case. *Management Science*, 40(3), 385–404.
- Gülpınar, N., & Rustem, B. (2007). Worst-case robust decisions for multi-period mean-variance portfolio optimization. *European Journal of Operational Research*, 183(3), 981–1000.
- Hakansson, N. (1971). Multi-period mean-variance analysis: Toward a general theory of portfolio choice. *Journal of Finance*, 26(4), 857–884.
- Jana, P., Roy, T., & Mazumder, S. (2009). Multi-objective possibilistic model for portfolio selection with transaction cost. *Journal of Computational and Applied Mathematics*, 228(1), 188–196.
- Konno, H., & Wiyayanayake, A. (2001). Portfolio optimization problem under concave transaction costs and minimal transaction unit constraints. *Mathematical Programming*, 89(2), 233–250.
- Kozhan, R., & Schmid, W. (2009). Asset allocation with distorted beliefs and transaction costs. *European Journal of Operational Research*, 194(1), 236–249.
- Kuan, C., Yeh, J., & Hsu, Y. (2009). Assessing value at risk with care, the conditional autoregressive expectile models. *Journal of Econometrics*, 150(2), 261–270.
- Li, D., Chan, T., & Ng, W. (1998). Safety-first dynamic portfolio selection. *Dynamics of Continuous, Discrete and Impulsive Systems*, 4(4), 585–600.
- Li, D., & Ng, W. (2000). Optimal dynamic portfolio selection: Multiperiod mean-variance formulation. *Mathematical Finance*, 10(3), 387–406.
- Liu, H. (2004). Optimal consumption and investment with transaction costs and multiple risky assets. *The Journal of Finance*, 59(1), 289–338.
- Lobo, M., Fazel, M., & Boyd, S. (2007). Portfolio optimization with linear and fixed transaction costs. *Annals of Operations Research*, 152(1), 341–365.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- Marzban, S., Mahootchi, M., & Khamseh, A. A. (2013). Developing a multi-period robust optimization model considering american style options. *Annals of Operations Research*, doi:10.1007/s10479-013-1461-x.
- McFarland, H. (1988). Evaluating q as an alternative to the rate of return in measuring profitability. *The Review of Economics and Statistics*, 75(4), 614–622.
- Merton, R. (1971). Optimum consumption and portfolio rules in a continuous-time model. *Journal of Economic Theory*, 3(4), 373–413.
- Osorio, M. A., Gülpınar, N., & Rustem, B. (2008). A general framework for multistage mean-variance post-tax optimization. *Annals of Operations Research*, 157(1), 3–23.
- Pelsser, A., & Vorst, T. (1996). Transaction costs and efficiency of portfolio strategies. *European Journal of Operational Research*, 91(2), 250–263.
- Perold, A. (1984). Large-scale portfolio optimization. *Management Science*, 30(10), 1143–1160.
- Samuelson, P. (1969). Lifetime portfolio selection by dynamic stochastic programming. *The Review of Economics and Statistics*, 51(3), 239–246.
- Shalit, H., & Yitzhaki, S. (2002). Estimating beta. *Review of Quantitative Finance and Accounting*, 18(2), 95–118.
- Steinbach, M. C. (2001). Markowitz revisited: Mean-variance models in financial portfolio analysis. *SIAM Review*, 43(1), 31–85.
- Wagner, W. (2010). Diversification at financial institutions and systemic crises. *Journal of Financial Intermediation*, 19(3), 373–386.
- Zhang, W., Zhang, X., & Xu, W. (2010). A risk tolerance model for portfolio adjusting problem with transaction costs based on possibilistic moments. *Insurance Mathematics and Economics*, 46(3), 493–499.

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